

CHAPTER 1

INTRODUCTION

Estimating the value of residential properties is an important task in the housing market, as it directly affects buyers, sellers, and real estate developers. The price of a house does not depend on a single factor but is influenced by several elements such as floor area, location, number of rooms, construction year, and overall quality of the building. Due to the involvement of many variables, traditional price estimation methods often fail to provide accurate results. With the availability of large amounts of data and improvements in computational techniques, machine learning has emerged as a reliable approach for analyzing property data and predicting house prices more accurately.

The work focuses on building a data-driven model that can estimate house prices using historical housing information. The dataset used in this work contains a mix of numerical and categorical attributes that describe different aspects of residential properties. Before model development, the data was processed to remove inconsistencies, manage missing entries, and transform categorical attributes into a suitable numerical format. Data visualization and statistical analysis were carried out to gain insights into how different features influence housing prices and to identify important relationships within the dataset.

To perform price estimation, a regression-based machine learning approach was adopted using the Linear Regression algorithm. The data was divided into separate training and testing sets to check how well the model performs on unseen data. The effectiveness of the model was measured using evaluation techniques such as error calculation and goodness-of-fit analysis. The comparison between actual values and predicted outputs helped in understanding the strengths and limitations of the model. Through this work, practical knowledge was gained in data handling, model building, and applying machine learning techniques to solve real-world problems related to housing price prediction.

CHAPTER 2

COMPAY PROFILE

Company Name: Infyro Solutions

Location: Trichy, Tamil Nadu, India

About the Company

Infyro Solutions is a technology-driven organization that provides innovative IT solutions and digital services to businesses across various domains. The company focuses on delivering high-quality software development, data-driven solutions, and modern technology services tailored to meet industry requirements. Infyro Solutions emphasizes practical learning by offering real-time projects and internship programs that help students and professionals gain hands-on technical experience. The organization is committed to bridging the gap between academic knowledge and industry expectations by promoting skill-oriented training and real-world project exposure.

Vision

- To empower individuals with practical technical skills aligned with current industry standards.
- To deliver reliable and innovative technology solutions for business transformation.
- To foster a learning-driven environment that supports growth in emerging technologies.

Mission

- To provide hands-on training through real-time projects and industry-relevant internships.
- To support continuous learning and professional development.

Key Services

- Data Analytics and Machine Learning Services
- Internship and Industrial Training Programs
- Real-Time Project Guidance.

CHAPTER 3

OBJECTIVES

- To design a system that can estimate residential property prices using machine learning techniques.
- To study how different housing features influence the overall value of a property.
- To collect and organize housing data for effective analysis and model development.
- To clean and prepare the dataset by handling missing values and inconsistencies.
- To transform non-numeric property information into a numerical format suitable for modeling.
- To explore patterns and relationships within the data using basic visual analysis methods.
- To apply a regression-based algorithm for predicting continuous price values.
- To assess the accuracy of the prediction model using appropriate performance measures.
- To compare actual property prices with predicted values to evaluate model reliability.
- To gain practical experience in applying machine learning concepts to real-world housing data.

CHAPTER 4

PLATFORM LEARNED

During the internship, several technical platforms and tools were used to develop the House Price Prediction system. The work was implemented using Python as the primary programming language due to its strong support for data analysis and machine learning tasks. All development and experimentation were carried out in Jupyter Notebook, which provided an interactive environment for executing code step-by-step, analyzing results, and visualizing outputs.

The following platforms and tools were learned and applied during the task:

- **Python Programming:** Used for data preprocessing, feature encoding, regression model implementation, and performance evaluation.
- **Pandas and NumPy:** Utilized for loading the dataset, handling missing values, performing data cleaning, and carrying out numerical computations.
- **Scikit-Learn:** Applied to implement the Linear Regression model, split the dataset into training and testing sets, and evaluate model performance using regression metrics such as MAE, MSE, RMSE, and R^2 score.
- **Matplotlib:** Used to visualize data relationships, correlation heatmaps, and comparison plots between actual and predicted house prices.
- **Machine Learning Concepts:** Gained practical understanding of supervised learning, regression techniques, feature selection, model training, and evaluation methods.

Overall, these tools and technologies supported the successful development and evaluation of the House Price Prediction model and enhanced practical knowledge of machine learning workflows.

CHAPTER 5

INTERNSHIP TASK

5.1. Data Collection

This task describes the data collection activity carried out for the House Price Prediction. The objective of this task was to gather reliable and well-structured housing data that captures the key factors influencing property prices.

5.2. Data Collection Overview

The dataset used in this work was collected from real estate-related data sources that record property details at the time of listing or sale. The data includes residential properties from different area categories such as urban, suburban, rural, and downtown locations, ensuring diversity in housing characteristics. All collected data was organized in a structured tabular format and stored in CSV files, making it suitable for analysis using Python-based tools and machine learning algorithms.

5.3. Dataset Description

The complete dataset contains 2000 records and 10 attributes, with each record representing an individual house. It is a structured dataset suitable for house price prediction, feature analysis, and regression-based modeling.

5.4. Dataset Structure

Each record in the dataset provides information related to:

- Physical characteristics of the house
- Construction year and condition
- Location category
- Availability of amenities
- Market price of the property

This structured format supports effective data preprocessing and model training.

5.5. Description of Variables

1. Property Size and Layout

- Area – Total built-up area of the house (in square feet)
- Bedrooms – Number of bedrooms
- Bathrooms – Number of bathrooms
- Floors – Number of floors

2. Construction Details

- YearBuilt – Year of construction
- Condition – Overall condition of the property (Excellent, Good, Fair, Poor)

3. Location Information

- Location – Area type such as Urban, Suburban, Rural, or Downtown

4. Amenities

- Garage – Indicates the availability of a garage (Yes/No)

5. Price Information

- Price – Target variable representing the house price

5.6. Nature of the Data

The dataset includes:

- Numerical variables: Area, Bedrooms, Bathrooms, Floors, YearBuilt, Price
- Categorical variables: Location, Condition, Garage

5.7. Outcome of the Data Collection Task

Through this data collection task, a clean and comprehensive housing dataset was successfully prepared. The structured data enabled effective analysis and served as a strong foundation for building and evaluating the house price prediction model.

5.8. System Workflow

After the data collection process, the prepared dataset is loaded into the Python environment for further processing and analysis. The program begins by importing the required libraries and reading the dataset from a CSV file. Initial checks are performed to understand the structure of the data, after which missing values are removed to maintain data quality and consistency. Categorical features such as location, property condition, and garage availability are converted into numerical values using encoding techniques so they can be used in machine learning models. Correlation analysis is then performed to study the relationship between different input features and house prices.

The dataset is separated into input variables and the target variable and then divided into training and testing sets. A Linear Regression model is trained using the training data and applied to predict house prices on the testing data. The performance of the model is evaluated using error metrics and graphical comparison between actual and predicted values. Finally, the trained model is used to predict the price of a sample house, demonstrating the practical working of the house price prediction system.

SOURCE CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
df = pd.read_csv("/House Price Prediction Dataset.csv")
print(df.head())
print(df.info())
df.dropna(inplace=True)
df = pd.get_dummies(df, drop_first=True)
corr_matrix = df.corr()
plt.figure(figsize=(10,6))
plt.imshow(corr_matrix, cmap='coolwarm', aspect='auto')
plt.colorbar()
plt.title("Correlation Heatmap")
```

```

plt.xticks(range(len(corr_matrix.columns)), corr_matrix.columns, rotation=90)
plt.yticks(range(len(corr_matrix.columns)), corr_matrix.columns)
plt.tight_layout()
plt.show()
X = df.drop("Price", axis=1)
y = df["Price"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R2 :", r2)
result = pd.DataFrame({
    "Actual Price": y_test,
    "Predicted Price": y_pred })
print(result.head())
plt.figure(figsize=(6,6))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Prices")
plt.show()
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_}).sort_values(by="Coefficient", ascending=False)
plt.figure(figsize=(8,6))
plt.barh(coefficients["Feature"], coefficients["Coefficient"])
plt.xlabel("Coefficient Value")

```



```
plt.title("Feature Importance")
plt.gca().invert_yaxis()
plt.show()
new_house = X.iloc[0:1]
new_prediction = model.predict(new_house)
print(new_prediction[0])
```

OUTPUT:

Table 5.1. Sample Dataset

Sno	Id	Area	Bedrooms	Bathrooms	Floors	Yearbuilt	Location	Condition
0	1	1360	5	4	3	1970	Downtown	Excellent
1	2	4272	5	4	3	1958	Downtown	Excellent
2	3	3592	2	2	3	1938	Downtown	Good
3	4	966	4	2	2	1902	Suburban	Fair
4	5	4926	1	4	2	1975	Downtown	Fair

Sno	Garage	Price
0	No	149919
1	No	424998
2	No	266746
3	No	244020
4	No	636056

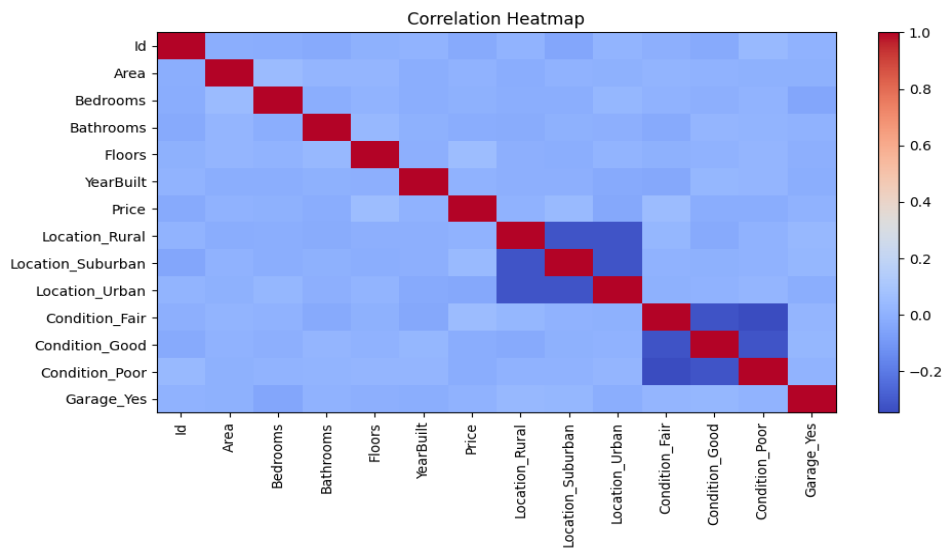


Figure 5.1. House Price HeatMap

===== **Model Performance** =====

MAE : 242867.44926338628

MSE : 78279764120.86243

RMSE: 279785.21069002635

R2 : -0.006181784611834162

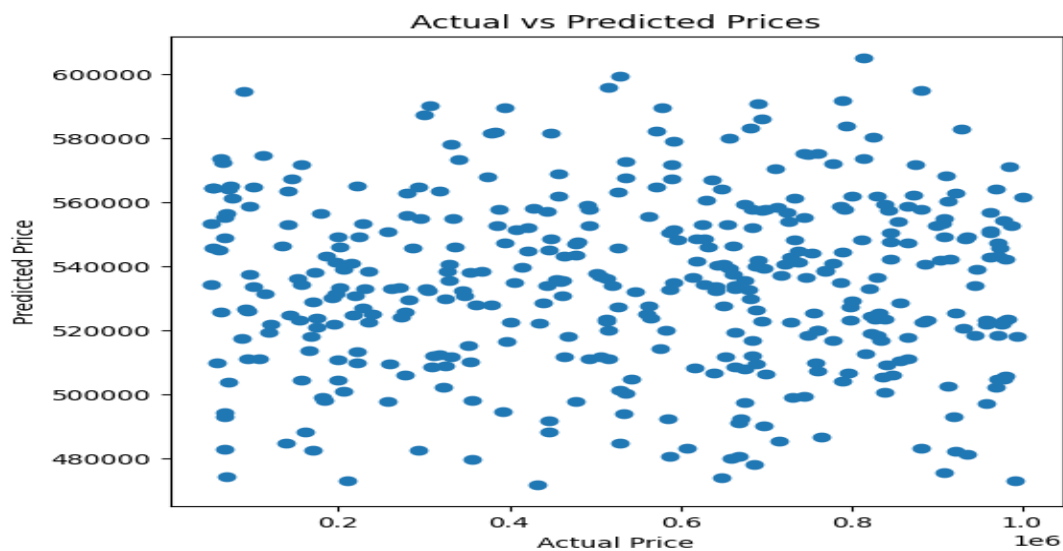


Figure 5.2 House Price Actual vs Predicted Prices

CHAPTER 6

LEARNING OUTCOMES

Completion of the internship task and the House Price Prediction task provided practical exposure to real-world data handling and fundamental machine learning concepts. A key learning outcome involved recognizing the importance of accurate and well-structured data collection, as data quality directly influences the performance of predictive models.

The work required to identifying and selecting relevant features for regression analysis, including property size, location, year of construction, and available amenities. This process enhanced the ability to analyze datasets and differentiate between numerical and categorical variables, which is essential for effective data preprocessing and feature encoding.

Technical proficiency in Python and data analysis libraries was strengthened through organizing large datasets, validating data consistency, and preparing data for machine learning models. The work also deepened understanding of how housing characteristics influence property prices and how such relationships can be modeled using regression techniques.

In addition, analytical and problem-solving skills were reinforced by addressing challenges such as data inconsistencies and feature relevance. Overall, the experience successfully bridged theoretical knowledge with practical implementation and enhanced readiness for real-world data science and machine learning applications.

CHAPTER 7

CONCLUSION

The House Price Prediction task effectively demonstrated the practical application of data collection and preparation techniques within a real-world problem domain. A structured dataset consisting of 2000 property records and 10 attributes was carefully compiled to support regression-based price prediction. The dataset captured essential housing features such as property area, number of bedrooms and bathrooms, construction year, location category, property condition, availability of garage facilities, and market price, ensuring comprehensive representation of factors influencing housing values. The data collection process played a vital role in establishing a strong foundation for subsequent stages, including data preprocessing, exploratory data analysis, and machine learning model development. Emphasis on data consistency, completeness, and relevance improved the overall quality and reliability of the dataset. This internship experience contributed significantly to the development of analytical reasoning and practical understanding of structured data handling. Exposure to real-world housing data strengthened familiarity with feature selection, variable classification, and preparation techniques required for predictive modeling. The knowledge and skills gained through this work are valuable for future applications in data analysis, machine learning, and regression-based predictive systems. Overall, the task successfully achieved its objectives and served as a meaningful learning experience in applied data science.